



# MongoDB

## Intro & Setting Up Atlas

*From documents to your first cluster*

# What is MongoDB?

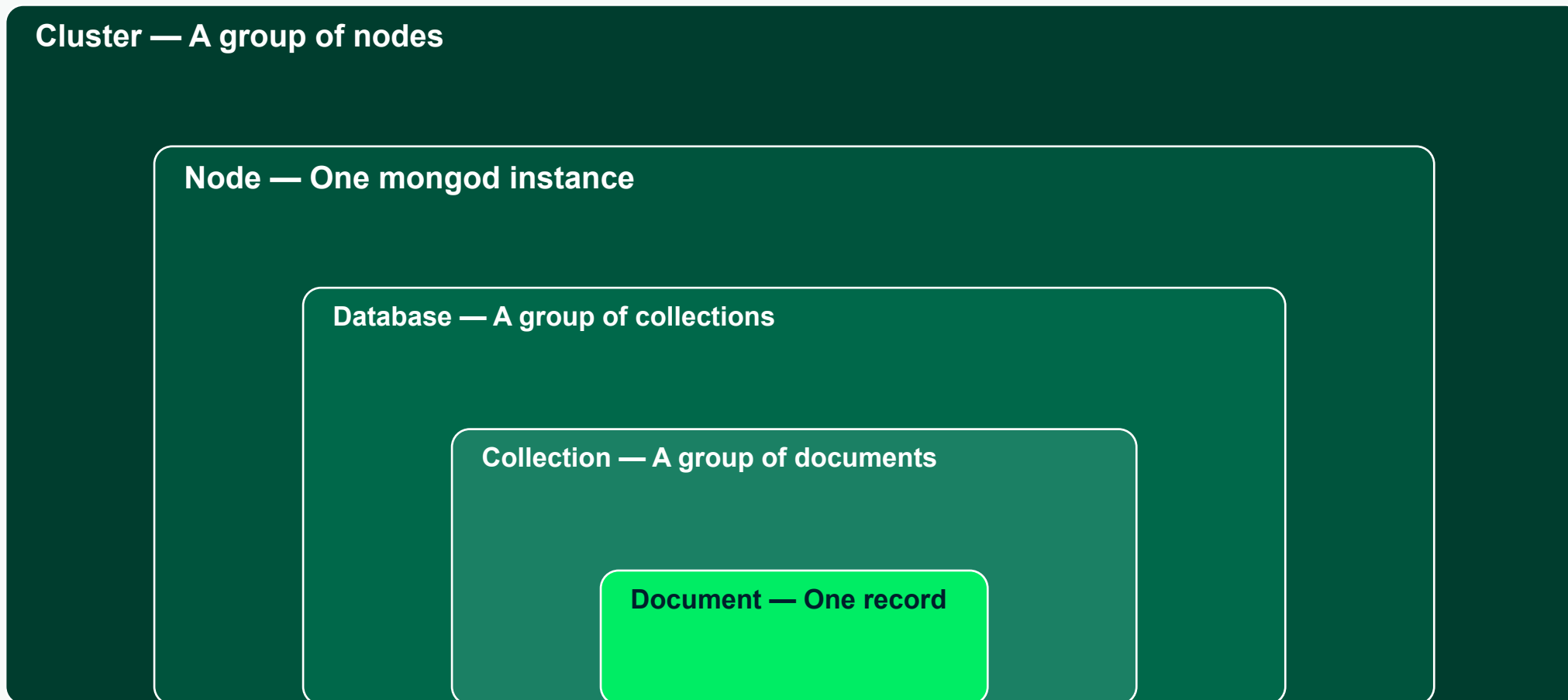
- A document-oriented, NoSQL database built for modern applications
- Instead of tables and rows, it stores JSON-like documents grouped into collections
- No fixed schema — documents in the same collection can look different from each other

```
{  
  name: "Priya",  
  branch: "CS",  
  cgpa: 8.4  
}
```

*A single document*

*Remember “What is a Database” from earlier? MongoDB is one specific, modern answer to that question.*

# The MongoDB Hierarchy



# JSON vs BSON

## JSON

*Human-readable text format*

```
{  
  "name": "Priya",  
  "age": 22  
}
```

## BSON

*Binary format — what MongoDB actually stores*

```
\x16\x00\x00\x00  
\x02name\x00...Priya\x00  
\x10age\x00\x16\x00\x00\x00  
(binary bytes)
```

### **BSON adds data types plain JSON doesn't have:**

- Date
- ObjectId
- Binary Data
- Int32 / Int64 / Decimal128

# Why BSON?

## Lightweight & Fast

Compact binary encoding — quick to traverse and parse

## Richer Data Types

Native support for dates, ObjectIds, and binary data

## Efficient Network Transfer

Smaller, faster to move between client and server

## Engine-Optimized

Built for how MongoDB actually stores and indexes data internally

# The Document Model

*“Data that is accessed together should be stored together.”*

**16 MB**

Max document size

**100**

Max levels of nesting

# One Document, Many Data Types

```
{  
  _id: ObjectId("665f...0a11"),  
  name: "Priya Sharma",  
  age: 22,  
  active: true,  
  joined: ISODate("2024-01-15"),  
  address: { city: "Pune" },  
  hobbies: ["chess", "coding"]  
}
```

ObjectId

String

Number

Boolean

Date

Object

Array

# MongoDB Data Types at a Glance

## String

Text, must be valid UTF-8

## Number

Int32, Int64, Double, Decimal128

## Boolean

true or false

## Object

A nested/embedded document

## Array

An ordered list of values

## Date

Stored as ms since Unix epoch

## Null

Explicit absence of a value

## ObjectId

Auto-generated unique document id

# Data Relationships

- Entities — the “things” in your app; usually one document per entity
- Attributes — the characteristics of an entity, stored as field-value pairs
- Relationships between entities come in a few standard shapes:

**1 – 1**

One student, one ID card

**1 – Many**

One student, many enrollments

**Many – Many**

Students and courses

# Embedding vs Referencing

## Embedded

```
{
  name: "Priya",
  courses: [
    { name: "DBMS", marks: 88 },
    { name: "OS", marks: 79 }
  ]
}
```

*One read gets everything*

## Referenced

```
// students
{ _id: 1, name: "Priya" }
```

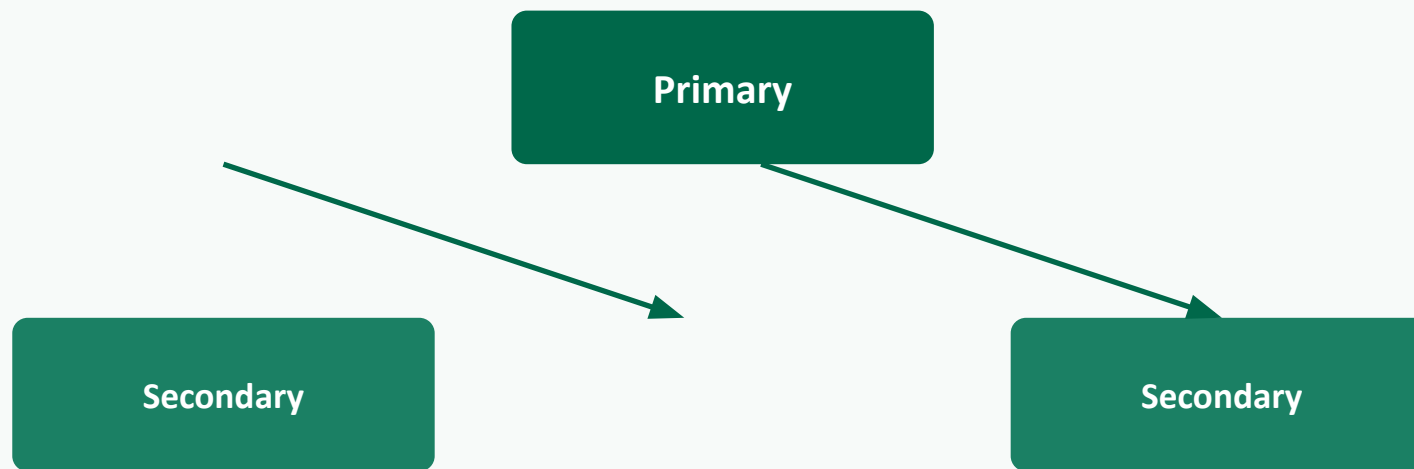
```
// enrollments
{ student_id: 1, course: "DBMS", marks: 88 }

{ student_id: 1, course: "OS", marks: 79 }
```

*Linked by student\_id — like a join*

*Embed when data is small and always read together. Reference when it's large, shared, or updated independently.*

# Replication — A Quick Preview

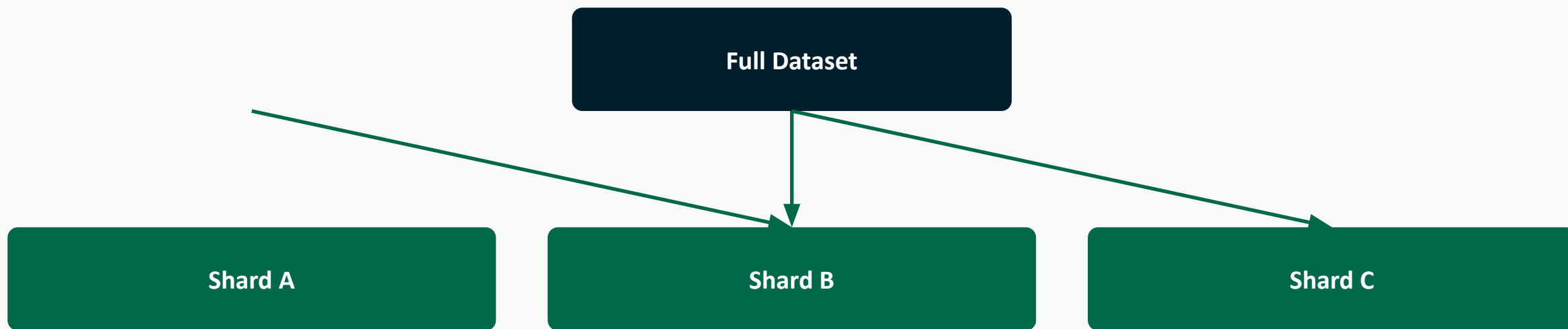


Same data, copied across multiple nodes

Why: availability — if one node goes down, the others keep serving.

*We'll go deep into why this matters from a systems angle shortly →*

# Sharding — A Quick Preview



Data split into pieces across multiple nodes

Why: scale — no single machine has to hold or process it all.

*More on this coming up →*

# What is MongoDB Atlas?

- A cloud-managed, Database-as-a-Service (DBaaS) version of MongoDB
- No manual server setup — MongoDB handles infrastructure, backups, scaling & security
- Runs on AWS, GCP, or Azure — your choice of provider and region

**Atlas is more than just a database:**

Atlas Search

Vector Search

Atlas Charts

Stream Processing

# Choosing a Cluster Tier

## M0

**Free / Shared**

Learning & small projects. This is what we use today.

## M2

**Serverless**

Pay only for what you use. Good for light, variable traffic.

## M10

**Dedicated**

Production workloads. Supports sharding & multi-region.

# Setting Up Your Cluster

1

## Create Account

cloud.mongodb.com — email,  
Google, or GitHub

2

## Build M0 Cluster

Pick any provider + nearest  
region

3

## Security Quickstart

Create a database user —  
username & password

4

## Network Access

Add your IP (or allow from  
anywhere for class use)

5

## Connect

Get your connection string —  
you're ready

# But why does MongoDB actually look and work this way?

*A big part of the answer: it's built as a distributed database from the ground up.*

Let's find out what that really means →